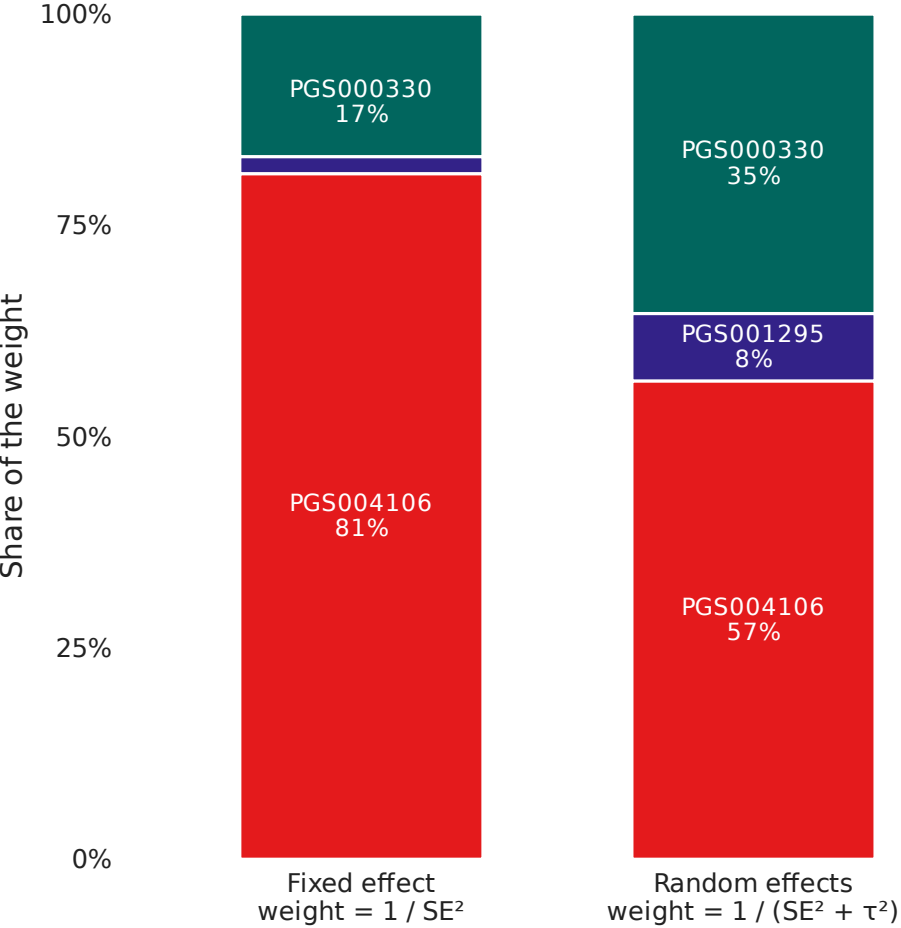


3. Random effects on these three scores adds a small τ^2 and keeps the deficit

I^2 near 51 means about half of Q is scatter beyond the sampling error expected for three estimates.
 τ^2 is small next to the variance of the most precise score, so the weights shift and the conclusion stays a deficit.

Who holds the weight



Three independent ΔAUC values

$$Q = \sum w (\Delta - \Delta_{IW})^2 = 4.08$$

$$df = k - 1 = 2$$

$$\tau^2 = \max(0, (Q - df) / C) = 0.00007$$

$$I^2 = \max(0, (Q - df) / Q) \times 100 = 51.0$$

C is the usual DerSimonian-Laird denominator,
 $\text{sum}(w) - \text{sum}(w^2) / \text{sum}(w)$.

τ^2 is added to every SE^2 before the weights are rebuilt. The pooled ΔAUC moves only because those weights move.

Pooled random-effects $\Delta AUC = -0.013$

95% interval -0.027 to -0.000

Q and I^2 are computed from the fixed-effect weights, then τ^2 rebuilds the weights. The same arithmetic, on the same three ΔAUC values, is what the review figure draws in green